

Java Full Stack Development Internship

Duration: 1 Month (22–23 Working Days | 1 Hour/Day) + 1 month Project

Week 1: Core Java Fundamentals

Day	Topics	Hands-on
Day 1	Program Orientation, Introduction to Full Stack Development	Install Java and IDE
Day 2	Variables, Data Types, Operators, Input/Output	Build a simple calculator program
Day 3	Conditional Statements and Loops	Pattern programs and menu-driven application
Day 4	Arrays and Strings	Array operations and string manipulation
Day 5	Methods, Return Types, Constructors.	Utility methods implementation

Week 2: Object-Oriented Programming

Day	Topics	Hands-on
Day 6	Classes and Objects	Create a Student Management class
Day 7	Encapsulation and Abstraction	Banking application example
Day 8	Inheritance (Types), Polymorphism (Method Overloading and Overriding)	Employee hierarchy implementation
Day 9	Interfaces and Exception Handling	Custom exception examples
Day 10	Multithreading Basics	Create and execute threads

Week 3: JDBC & Development Tools

Day	Topics	Hands-on
Day 11	JDBC Architecture, JDBC Drivers, Establishing Database Connections	Connect Java application to MySQL
Day 12	JDBC CRUD Operations using <code>PreparedStatement</code>	Perform CRUD operations
Day 13	Git, GitHub Basics, Database Connectivity Recap	Create repository and push code
Day 14	Introduction to React, React Project Structure	Create first React application

Week 4: React Fundamentals

Day	Topics	Hands-on
Day 15	JSX and Functional Components	Build reusable components
Day 16	Props and State (<code>useState</code>)	Create a counter application
Day 17	Hooks (<code>useEffect</code>)	Fetch and display data
Day 18	Fetch API, Axios, JSON Handling	Consume sample APIs
Day 19	API Integration and GET Requests	Display backend data in React

Week 5: React API Integration

Day	Topics	Hands-on
Day 20	POST, PUT, DELETE Requests	Submit forms and update records
Day 21	Connecting React with Backend APIs, Error Handling	Complete frontend-backend integration
Day 22	Recap, Best Practices, Project Phase Overview	End-to-end application flow demonstration

Project Phase

Duration: 1 Month

Project Requirements

The application should implement:

- Java Core Concepts
- JDBC Connectivity
- CRUD Operations
- React-based User Interface
- REST API Integration
- Git and GitHub Version Control

Deliverables

- 1 Final Project
- GitHub Repository Link
- Project Report
- PPT Presentation (8–10 Slides)